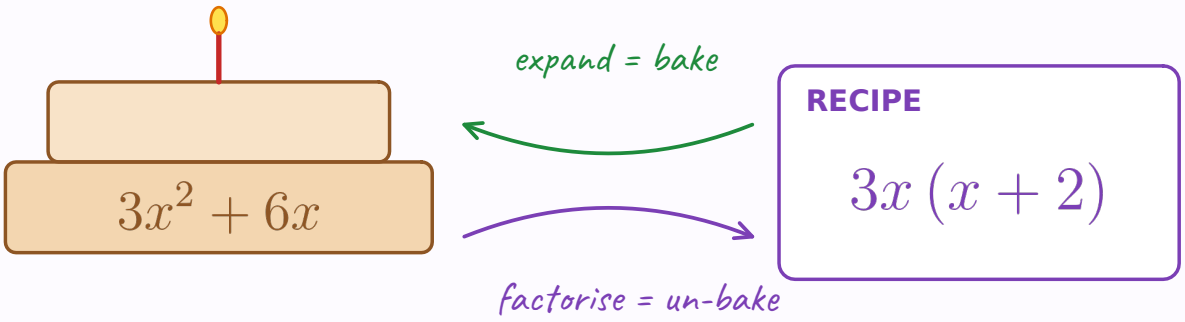
Chapter 12

Factorisation

un-baking: reading the recipe back off the cake



In Chapter 11 you **expanded**: brackets in, long expression out. **Factorising** is the same trip backwards. You're handed the long expression (the cake) and asked for the brackets that made it (the recipe).

It's harder in this direction, the way un-baking is harder than baking, so every shape gets a **rule stated in a box first**: take out common factors first; for three terms, find two numbers that **add** to one coefficient and **multiply** to another; for two terms, look for squares. A formula (the grid formula) is kept as a back-up that always finds the numbers, a map says which rule to use, and a summary page lists them all. Every answer can be checked by baking it again: expand it and see if you get back where you started.

Goes with **Lecture 2** (factorising, difference of two squares, perfect squares, cubes).

What "factor" means (numbers first)

A **factor** is something you multiply by. $12 = 3 \times 4$, so 3 and 4 are factors of 12 (Chapter 1). **Factorising** means rewriting something **as a multiplication**.

Try it with a sum of numbers: $6 + 9$. Both numbers are in the 3 times table:

$$6 + 9 = 3 \times 2 + 3 \times 3$$ ← *split each into 3 × something*

$$= 3 \times (2 + 3)$$ ← *the 3 is shared, so pull it out front*

$$= 3 \times 5 = 15$$ ← *same as 6 + 9 ✓*

That's all factorising is: spot what every term **shares**, write it once outside a bracket, and put what's **left over** inside. Your slides do the same with fractions:

$$\frac{2}{7} + \frac{2}{9} = 2\left(\frac{1}{7} + \frac{1}{9}\right)$$ ← *each term is 2 × something*

Common factors: break every term into atoms

PROCEDURE: taking out a common factor (always the first step)

1. Find the **highest common factor** of every term: the biggest number that divides all the numbers, times each letter to the **lowest** power it appears in every term.
2. Write it once outside a bracket. Inside, write what is left of each term (each term divided by it).
3. Check: the terms inside the bracket share nothing more, and expanding gives the question back.

Lecture 2 example: $3x^2 + 6x$. Break each term into the smallest things multiplied together (its "atoms"). Numbers split into primes, and x^2 splits into $x \times x$:

$$3x^2 = 3 \times x \times x$$

$$6x = 2 \times 3 \times x$$

highlighted = in BOTH rows (3 and x)

shared atoms go outside: 3 × x = 3x. left over: x (top row) and 2 (bottom row)

$$3x^2 + 6x = 3x\,(x + 2)$$ ← *outside: shared; inside: leftovers*

Bake it again to check: $3x \times x = 3x^2$ and $3x \times 2 = 6x$ ✓. And with a number, $x = 2$: $12 + 12 = 24$ and $6 \times 4 = 24$ ✓.

Take out the biggest shared thing. $3x^2 + 6x = 3(x^2 + 2x)$ is true, but the bracket still has a shared x inside, so you're not finished. Keep going until the terms in the bracket share nothing.

Three terms, same idea (your slides): $3x + 4qx + 12wx$. Every term has an x atom; the numbers 3, 4, 12 share nothing (4 isn't a multiple of 3). So only x comes out:

$$3x + 4qx + 12wx = x\,(3 + 4q + 12w)$$ ← *x × 3, x × 4q, x × 12w*

Why bother? Lecture 2 asks you to make x the subject of $3x + xy = z$. The x is stuck in two places; taking it out glues the two copies into one, and then one division finishes the job:

$$3x + xy = z \;\Rightarrow\; x(3+y) = z \;\Rightarrow\; x = \frac{z}{3+y}$$

one x, then divide

More common factors: chunks and pairs

Lecture 2: $(x+y)^3 + 7(x+y)^2$. It looks frightening, but $(x+y)$ appears whole every time. Cover it with a box and call it $\square$:

$$\square^3 + 7\,\square^2$$ ← $\square$ is just a name for $(x+y)$

$$= \square^2(\square + 7)$$ ← atoms: $\square^3 = \square \times \square \times \square$; shared: $\square^2$

$$= (x+y)^2(x+y+7)$$ ← take the box off again

Four terms: group them in pairs. The slide's example should read $x^2 + 2y + 2x^3 + 4xy$. Look at the last two terms: $2x^3 + 4xy$ share $2x$.

$$(x^2 + 2y) + (2x^3 + 4xy)$$ ← two pairs

$$= (x^2 + 2y) + 2x\,(x^2 + 2y)$$ ← the second pair: $2x \times (x^2 + 2y)$

$$= (x^2 + 2y)(1 + 2x)$$ ← $(x^2 + 2y)$ is shared; the first pair is $1 \times$ it

Slide check. The slide prints $2y^2$ instead of $2y$ and finishes with $(x^2 + 2y)(x^2 + 2y + 2x)$. Test with $x = y = 1$: the slide's start is $1 + 2 + 2 + 4 = 9$, but its answer is $3 \times 5 = 15$ ✗. The version above gives $3 \times 3 = 9$ ✓.

Three terms: find two numbers that add and multiply

RULE: factorising x² + bx + c (the number in front of x² is 1)

Find two numbers that **ADD to b** and **MULTIPLY to c** . Call them p and q.

Then $x^2 + bx + c = (x + p)(x + q)$.

Signs: if c is **positive**, both numbers have the same sign as b. If c is **negative**, one is positive and one is negative, and the bigger one (ignoring signs) takes the sign of b.

Search the **factor pairs of c** (the multiply condition is easier to list), and test each pair's sum.

Why it works: expand $(x + p)(x + q) = x^2 + px + qx + pq = x^2 + (p + q)x + pq$. The middle number is the **sum** and the last number is the **product**. Lecture 2 says it in words: "the 1 and 3 are each multiplied by x, giving $x + 3x = 4x$; the 1 and 3 are multiplied together to give 3."

Example 1: $x^2 + 7x + 12$. Need: add to 7, multiply to 12.

$$12 = 1 \times 12, \quad 2 \times 6, \quad 3 \times 4$$ ← factor pairs of 12

$$1 + 12 = 13, \quad 2 + 6 = 8, \quad 3 + 4 = 7 \checkmark$$ ← test each sum: 3 and 4 work

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Example 2: $x^2 - x - 6$. Here $b = -1$ and $c = -6$: the numbers have opposite signs (c is negative) and the bigger one is negative (b is negative).

$$-6 = 1 \times (-6), \quad 2 \times (-3), \quad (-1) \times 6, \quad (-2) \times 3$$

pairs that multiply to −6

$$1 - 6 = -5, \quad 2 - 3 = -1 \checkmark$$ ← 2 and −3 add to −1

$$x^2 - x - 6 = (x + 2)(x - 3)$$

Bake it to check: $(x + 2)(x - 3) = x^2 - 3x + 2x - 6 = x^2 - x - 6$ ✓.

PROCEDURE: factorising $ax^2 + bx + c$ when a is not 1: split the middle

1. Take out any common factor first. Then work out $a \times c$.
2. Find two numbers that **MULTIPLY to ac** and **ADD to b**.
3. **Split the middle term** bx into two x terms using those numbers.
4. **Factorise in pairs**: a common factor from the first two terms, and from the last two.
5. The two brackets now match: take that bracket out as a common factor.

Example 3: $6x^2 + 7x - 3$. So $a = 6$, $b = 7$, $c = -3$.

$$ac = 6 \times (-3) = -18 \quad \leftarrow \text{1. a × c}$$

$$9 \times (-2) = -18, \quad 9 + (-2) = 7 \quad \leftarrow \text{2. multiply to −18, add to 7}$$

$$6x^2 + 9x - 2x - 3 \quad \leftarrow \text{3. split 7x into 9x − 2x}$$

$$= 3x(2x + 3) - 1(2x + 3) \quad \leftarrow \text{4. factorise each pair}$$

$$= (2x + 3)(3x - 1) \quad \leftarrow \text{5. (2x + 3) is shared}$$

In step 4, take out −1 (not +1) from $-2x - 3$, so that the second bracket matches the first: $-2x - 3 = -1(2x + 3)$. If the brackets don't match, check the signs or swap the order of the two middle terms.

✗ Common mistakes

✗ $x^2 + 5x + 6 = (x + 6)(x + 1)$ ✓ $x^2 + 5x + 6 = (x + 2)(x + 3)$

6 and 1 multiply to 6 but add to 7: the numbers must do BOTH

✗ $x^2 - 2x - 8 = (x + 2)(x + 4)$ ✓ $x^2 - 2x - 8 = (x + 2)(x - 4)$

c is negative, so one number is negative; the bigger one takes the sign of b

✗ $2x^2 + 8x + 6 = (x + 1)(x + 3)$ ✓ $2x^2 + 8x + 6 = 2(x + 1)(x + 3)$

don't lose the common factor you took out

When the search is slow (big numbers, or no whole numbers work), the next section gives a formula that **computes** the two numbers directly instead of searching for them.

Another way for three terms: the grid formula

This is an **alternative** to the search. It finds the very same pair of numbers (they add to b and multiply to ac), but by formula, so it also works when the numbers aren't whole. Chapter 11 filled a grid from its edges. Factorising $ax^2 + bx + c$ runs it **backwards**: the boxes are the question, the edges are the answer, and the edges have a formula:

THE GRID FORMULA for $ax^2 + bx + c$

$\times$	ax	$+m$
x	ax^2	$m\,x$
$+\dfrac{n}{a}$	$n\,x$	c

box = side edge × top edge

1. $\Delta = b^2 - 4ac$
2. $m,\ n = \dfrac{b \pm \sqrt{\Delta}}{2}$
3. $\left(x + \dfrac{n}{a}\right)(ax + m)$

Δ is the Greek D, "delta". The two numbers m and n are **the pair**. (Take out any common factor first.)

Worked example: $2x^2 + 7x + 3$. Read off the letters: $a = 2$, $b = 7$, $c = 3$.

$$\Delta = 7^2 - 4 \times 2 \times 3 = 49 - 24 = 25 \quad \longleftarrow \text{1. } 25 = 5^2$$

$$m = \frac{7+5}{2} = 6, \qquad n = \frac{7-5}{2} = 1 \quad \longleftarrow \text{2. the pair}$$

3. Put the numbers into the grid, in the places the letters were:

$\times$	$2x$	$+6$
x	$2x^2$	$+6x$
$+\dfrac{1}{2}$	$+x$	$+3$

Top edges: $ax = 2x$ and $m = 6$.

Side edges: x and $\frac{n}{a} = \frac{1}{2}$.

Boxes: $2x \times x = 2x^2$, then $6x + x = 7x$, then $\frac{1}{2} \times 6 = 3$. That's the question ✓

$$2x^2 + 7x + 3 = \left(x + \tfrac{1}{2}\right)(2x + 6) \quad \longleftarrow \text{read the edges}$$

$$= (2x + 1)(x + 3) \quad \longleftarrow \text{clear the fraction: } \times 2 \text{ and } \div 2$$

Doubling one bracket and halving the other changes nothing, like $3 \times 4 = 6 \times 2$. **Bake it to check:** $(2x+1)(x+3) = 2x^2+6x+x+3 = 2x^2+7x+3$ ✓.

When $a = 1$ there's no fraction: the answer is just $(x+n)(x+m)$. For $x^2+7x+12$, $\Delta = 49-48 = 1$ and the pair is $\frac{7\pm1}{2} = 4, 3$, so it's $(x+3)(x+4)$.

> ✎ **YOUR GO (pencil first)**
>
> *$3x^2 + 10x + 8$. Work out Δ, then the pair, then put them into the grid formula and clear the fraction.*

Answer: $\Delta = 100-96 = 4$, pair $\frac{10\pm2}{2} = 6, 4$. Top edges $3x$ and 6, side edges x and $\frac{4}{3}$. So $\left(x+\frac{4}{3}\right)(3x+6)$; × 3 and ÷ 3 gives $(3x+4)(x+2)$.

Why the grid formula always works

Multiply the card's edges back out, box by box. Every box comes out as the question, whatever a, b and c are:

$$x \times ax = ax^2 \longleftarrow \text{top-left ✓}$$

$$mx + nx = (m+n)\,x = bx \longleftarrow \text{middle ✓ : the } \sqrt{\Delta} \text{ parts cancel}$$

$$\frac{n}{a} \times m = \frac{mn}{a} = \frac{ac}{a} = c \longleftarrow \text{bottom-right ✓ (needs } mn = ac\text{)}$$

And $mn = ac$ is where Δ comes in. Multiply the pair; the top is a difference of two squares (Chapter 11):

$$mn = \frac{(b+\sqrt{\Delta})(b-\sqrt{\Delta})}{4} = \frac{b^2-\Delta}{4} = \frac{4ac}{4} = ac$$

because $\Delta = b^2 - 4ac$

That's the whole reason Δ is $b^2 - 4ac$: it's the number that makes the corner come out as c. Chapter 15 sets each bracket to zero, $x = -\frac{n}{a}$ or $x = -\frac{m}{a}$, and that turns into the quadratic formula.

What Δ tells you before you start

Step 1 sorts every quadratic before you touch the grid:

quadratic	Δ = b² − 4ac	√Δ	pair	
$2x^2 + 7x + 3$	$49 - 24 = 25$	5	$6,\ 1$	*a square: whole brackets*
$x^2 + 4x + 4$	$16 - 16 = 0$	0	$2,\ 2$	*zero: (x + 2) twice*
$x^2 + 4x + 2$	$16 - 8 = 8$	$\sqrt{8}$	$2 \pm \sqrt{2}$	*not a square: square roots*
$6x^2 + 4x + 3$	$16 - 72 = -56$	–	none	*negative: no brackets*

Not a square: the grid formula still works; the edges just contain square roots. There are no whole-number brackets, so we say it **doesn't factorise with whole numbers** (the add-and-multiply search finds nothing). With square roots allowed, it does:

$$x^2 + 4x + 2 = (x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

pair 2 ± √2, a = 1

Negative: no number squared is negative, so $\sqrt{\Delta}$ doesn't exist: no pair, **no brackets at all**. You haven't made a mistake; there's nothing to find. Exam questions are picked so that Δ is a perfect square. Chapter 15 shows the three cases as graphs that cross the x-axis twice, touch it once, or miss it.

Minus signs: keep each sign with its number

$6x^2 - x - 2$: $a = 6$, $b = -1$, $c = -2$. (Split the middle would search for two numbers that multiply to $ac = -12$ and add to -1: that's 3 and -4.) The grid formula's steps:

$$\Delta = (-1)^2 - 4 \times 6 \times (-2) = 1 + 48 = 49$$ ← 1. 49 = 7²

$$m = \frac{-1+7}{2} = 3, \qquad n = \frac{-1-7}{2} = -4$$ ← 2. the pair

$\times$	$6x$	$+3$
x	$6x^2$	$+3x$
$-\frac{2}{3}$	$-4x$	-2

3. **Top:** $6x$ and $+3$. **Side:** x and $\frac{n}{a} = \frac{-4}{6} = -\frac{2}{3}$.

Boxes: $3x - 4x = -x$ and $-\frac{2}{3} \times 3 = -2$ ✓. You never chose a minus sign: the formula handed it to you.

$$6x^2 - x - 2 = \left(x - \tfrac{2}{3}\right)(6x + 3)$$ ← read the edges

$$= (3x - 2)(2x + 1)$$ ← clear the fraction: × 3 and ÷ 3

Same answer, different look. $(3x - 2)(2x + 1)$ and $(2x + 1)(3x - 2)$ are the same, like $3 \times 4 = 4 \times 3$. Lecture 2 writes $2x^2 + 5x + 3 = 2(x + 1.5)(x + 1)$ and says it "isn't the only solution": moving the 2 into the first bracket gives $(2x + 3)(x + 1)$, the whole-number version an examiner expects.

The grid's two rows, written in a line, are exactly **split the middle** from the start of this part (the pair 6, 1 multiplies to $ac = 6$ and adds to $b = 7$):

$$2x^2 + 7x + 3 = (2x^2 + 6x) + (x + 3)$$ ← the grid's two rows

$$= 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$$ ← each row's side edge, then the shared bracket

Difference of two squares: cut and slide

RULE: difference of two squares

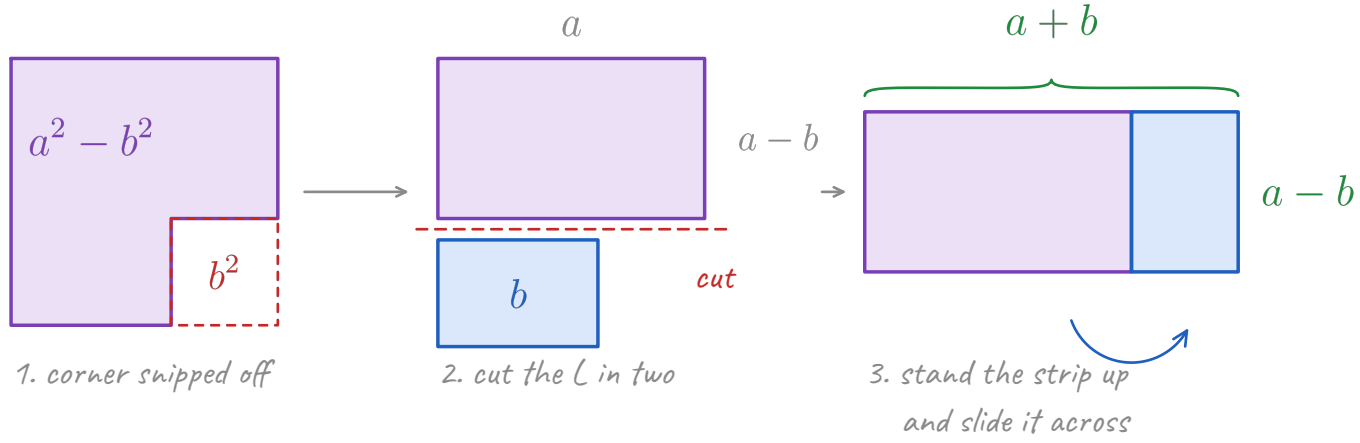
$a^2 - b^2 = (a - b)(a + b)$.

Use it when there are exactly **two** terms, a **minus** between them, and **each term is a square** (of a number, a letter, or both). Ask "a square of what?" to find a and b.

Take out a common factor first if there is one: $2x^2 - 18 = 2(x^2 - 9)$.

A sum of two squares, $a^2 + b^2$, does not factorise.

Why it's true: take a square of paper, a by a, and snip a small b by b square off one corner. What's left has area $a^2 - b^2$. Now watch:



Same paper, now a rectangle $(a + b)$ wide and $(a - b)$ tall:

$$a^2 - b^2 = (a - b)(a + b)$$

How to spot it: exactly **two** terms, a **minus** between them, and each term is a square. Then ask "a square of what?"

$x^2 - 9 = x^2 - 3^2 = (x - 3)(x + 3)$ ← 9 is 3 squared

$x^2 - 4y^2 = x^2 - (2y)^2 = (x - 2y)(x + 2y)$

$4y^2$ is $(2y)$ squared

It's also just the method with $b = 0$: for $x^2 - 9$, $\Delta = 0 + 36 = 36$ and the pair is ± 3. Spotting the two squares is simply faster. (The slide says "Expand: $x^2 - 4y^2$" but then factorises it; "factorise" is the right word.)

Party trick: $51 \times 49 = (50+1)(50-1) = 50^2 - 1^2 = 2499$, no calculator needed. **Warning:** $a^2 + b^2$ (a **plus**) does not factorise this way.

Perfect squares: a bracket times itself

RULE: perfect squares

$a^2 + 2ab + b^2 = (a+b)^2$ and $a^2 - 2ab + b^2 = (a-b)^2$.

How to spot it: the **first and last terms are squares** (a^2 and b^2, last term positive) and the **middle term is 2 × a × b**. The sign of the middle term goes in the bracket.

If the middle isn't $2ab$, it is not a perfect square: use the add-and-multiply rule instead.

It's the add-and-multiply rule with the two numbers equal: for $x^2 + 6x + 9$ the numbers are 3 and 3 (add to 6, multiply to 9), so both brackets are the same.

$$x^2 + 6x + 9 = x^2 + 2 \cdot x \cdot 3 + 3^2 = (x+3)^2$$

a = x, b = 3; middle 2·x·3 = 6x ✓

$$4x^2 - 20x + 25 = (2x)^2 - 2 \cdot 2x \cdot 5 + 5^2 = (2x-5)^2$$

a = 2x, b = 5; middle 20x ✓

Find the missing number. "What must k be for $x^2 + 10x + k$ to be a perfect square?" The bracket must be $(x+5)^2$, because $2 \times x \times 5 = 10x$. So:

$$k = 5^2 = 25$$

half of 10, squared: $(x+5)^2 = x^2 + 10x + 25$

✗ Common mistakes

✗ $x^2 - 9 = (x-3)^2$ ✓ $x^2 - 9 = (x-3)(x+3)$

$(x-3)^2 = x^2 - 6x + 9$: two terms with a minus is a difference of squares

✗ $x^2 + 9 = (x+3)(x-3)$ ✓ $x^2 + 9$ doesn't factorise

$(x+3)(x-3) = x^2 - 9$; a sum of squares has no brackets

✗ $x^2 + 4x + 16 = (x+4)^2$

the middle must be 2 × 4 × x = 8x

✓ $(x+4)^2 = x^2 + 8x + 16$

Cubes: a bigger grid

Lecture 2 lists two cube patterns. The same grid proves them, with three columns instead of two:

$$(a-b)(a^2+ab+b^2)$$

$\times$	a^2	$+ab$	$+b^2$
a	a^3	$+a^2b$	$+ab^2$
$-b$	$-a^2b$	$-ab^2$	$-b^3$

Same colour, opposite sign: the **blue** boxes cancel and the **red** boxes cancel. Only the two **green** corners are left: $a^3 - b^3$.

$$a^3 - b^3 = (a-b)(a^2+ab+b^2)$$ ← *a = 2, b = 1: 8 − 1 = 1 × 7*

Swap b for $-b$ everywhere and the same grid gives the plus version:

$$a^3 + b^3 = (a+b)(a^2-ab+b^2)$$ ← *a = 2, b = 1: 8 + 1 = 3 × 3*

Remembering the signs, "SOAP": the first bracket has the **S**ame sign as the question, the middle of the second bracket is **O**pposite, and the last term is **A**lways **P**ositive.

Using it: spot two cubes, find a and b, and fill in the pattern.

$$8x^3 - 27 = (2x)^3 - 3^3$$ ← *$8x^3$ is (2x) cubed, 27 is 3 cubed*

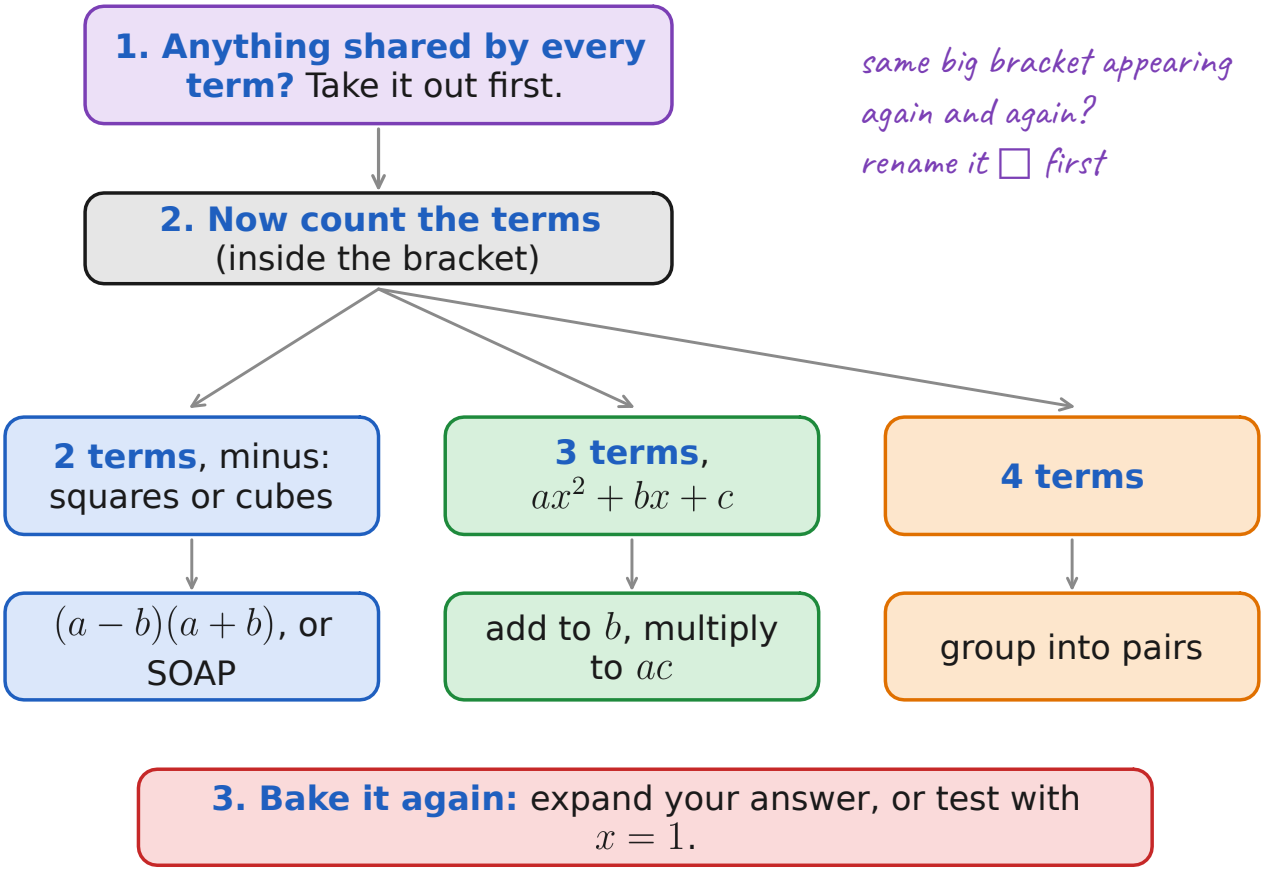
$$= (2x-3)(4x^2+6x+9)$$ ← *a = 2x, b = 3 in the minus pattern*

The map: which tool do I use?

PROCEDURE: factorising anything

1. **Always take out common factors first.** Look for a number, letter or bracket in every term.
2. Count the terms left in the bracket.
3. **Two terms:** two squares with a minus give $(a-b)(a+b)$; two cubes, plus **or** minus, use SOAP. A sum of two squares doesn't factorise.
4. **Three terms:** a perfect square $(a \pm b)^2$? If not: x^2+bx+c, find two numbers adding to b and multiplying to c; ax^2+bx+c, split the middle (multiply to ac, add to b). Stuck? Use the grid formula, or Δ to see if whole-number brackets exist at all.
5. **Four terms:** group into pairs and take out the shared bracket.
6. Check each bracket can't be factorised further, then expand to check the whole answer.

The same procedure as a map. Most questions need two stops: a common factor first, then one of the others.



Factorising rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Common factor first	take out the HCF of every term before anything else
x² + bx + c	$(x+p)(x+q): \quad p+q=b, \quad pq=c$ two numbers that **add** to b and **multiply** to c
ax² + bx + c (split the middle)	$pq=ac, \quad p+q=b: \quad ax^2+px+qx+c$ then factorise in pairs and take out the shared bracket
Grid formula (alternative)	$m,\, n=\dfrac{b\pm\sqrt{b^2-4ac}}{2}, \quad \left(x+\dfrac{n}{a}\right)(ax+m)$ computes the same pair; clear any fraction at the end
Does it factorise?	$\Delta=b^2-4ac$ perfect square: whole brackets; not a square: not with whole numbers; negative: not at all
Difference of two squares	$a^2-b^2=(a-b)(a+b)$ a^2+b^2 doesn't factorise
Perfect squares	$a^2\pm 2ab+b^2=(a\pm b)^2$ first and last square, middle $2ab$
Grouping (4 terms)	$ax+ay+bx+by=(a+b)(x+y)$ pairs, then the shared bracket
Difference of cubes	$a^3-b^3=(a-b)(a^2+ab+b^2)$
Sum of cubes	$a^3+b^3=(a+b)(a^2-ab+b^2)$ SOAP: Same, Opposite, Always Positive
Check	expand your answer: you must get the question back

Un-baking: your turn

Answers in grey. Bake each answer again (expand it) to check yourself.

Common factor

1. $4x + 12$ 2. $6x^2 - 9x$ 3. $5ab + 10a$ 4. $2x^3 + 4x^2 + 8x$

1. $4(x + 3)$ 2. $3x(2x - 3)$ 3. $5a(b + 2)$ 4. $2x(x^2 + 2x + 4)$

Three terms: the pair adds to b and multiplies to ac (Δ shows whether it exists)

5. $x^2 + 7x + 12$ 6. $x^2 - x - 20$ 7. $3x^2 + 6x - 24$
8. $2x^2 + 11x + 5$ 9. $6x^2 + 7x + 2$ 10. $4x^2 - 4x - 3$
11. $5x^2 + 13x - 6$

5. $\Delta = 1$, pair 4, 3: $(x + 3)(x + 4)$ 6. $\Delta = 81$, pair 4, −5: $(x - 5)(x + 4)$
7. take out 3 first; $\Delta = 36$, pair 4, −2: $3(x + 4)(x - 2)$ 8. $\Delta = 81$, pair 10, 1: $(2x + 1)(x + 5)$
9. $\Delta = 1$, pair 4, 3: $(2x + 1)(3x + 2)$ 10. $\Delta = 64$, pair 2, −6: $(2x - 3)(2x + 1)$
11. $\Delta = 289$, pair 15, −2: $(5x - 2)(x + 3)$

Whole, square roots, or nothing? (Δ first)

12. $x^2 + 6x + 8$ 13. $x^2 + 6x + 7$ 14. $x^2 + 2x + 3$

12. $\Delta = 4 = 2^2$: pair 4, 2, $(x + 2)(x + 4)$ 13. $\Delta = 8$: pair $3 \pm \sqrt{2}$, $(x + 3 - \sqrt{2})(x + 3 + \sqrt{2})$
14. $\Delta = -8$: no brackets

Difference of two squares

15. $x^2 - 25$ 16. $4x^2 - 49$ 17. $2x^2 - 18$
18. 97×103 (use the trick)

15. $(x - 5)(x + 5)$ 16. $(2x - 7)(2x + 7)$ 17. $2(x - 3)(x + 3)$ 18. $100^2 - 3^2 = 9991$

Grouping and chunks

19. $ax + ay + bx + by$
20. $(x - 1)^2 + 3(x - 1)$
21. Make x the subject: $ax - b = cx$

19. $(a + b)(x + y)$ 20. $(x - 1)(x + 2)$ 21. $x = b/(a - c)$

Cubes

22. $x^3 - 8$ 23. $27x^3 + 1$

22. $(x - 2)(x^2 + 2x + 4)$ 23. $(3x + 1)(9x^2 - 3x + 1)$

Find the unknown

24. $x^2 + kx + 12 = (x + 3)(x + p)$. Find p and k.
25. Find k so that $x^2 + 6x + k$ is a perfect square.
26. $x^2 - 5x + c = (x - 2)(x - q)$. Find q and c.
27. $(x + 4)$ is a factor of $x^2 + x + k$. Find k and the other factor.
28. Find the positive k so that $4x^2 + kx + 9$ is a perfect square.
29. $2x^2 + bx - 3 = (2x - 1)(x + 3)$. Find b.

24. $p = 4$, $k = 7$ ($3p = 12$, $k = 3 + p$) 25. $k = 9$ ($(6 \div 2)^2 = 9$)
26. $q = 3$, $c = 6$ ($2 + q = 5$, $c = 2q$) 27. $k = -12$, $(x - 3)$ ($4 + p = 1$, so $p = -3$; $k = 4p$)
28. $k = 12$ ($(2x + 3)^2$, middle $2 \times 2x \times 3$) 29. $b = 5$ (expand: $6x - x = 5x$)